

ACI 311.7-18

An ACI Standard

Specification for Inspection of Concrete Construction

Reported by ACI Committee 311



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Specification for Inspection of Concrete Construction

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Reported by ACI Committee 311

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This reference specification covers quality assurance inspection services for preplacement, placement, and post-placement of concrete construction. This reference specification can be made applicable to a particular construction project by citing it in the inspection services contract. The specifier shall supplement the provisions of this reference specification as needed by specifying individual project requirements in the inspection services contract. The materials, processes, quality control measures, and inspections described in this specification should be tested, monitored, or performed as applicable only by individuals holding the appropriate ACI Certifications or equivalent.

Keywords: accreditation; certification; concrete construction; inspection agency; placement; post-placement; preplacement; special inspection; specification.

(mandatory portion follows)

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ACI 311.7-18 supersedes ACI 311.7-14, became effective March 16, 2018, and was published April 2018.

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SECTION 1—GENERAL REQUIREMENTS

1.1—Scope

1.1.1 Services specified—This specification sets the minimum requirements for quality assurance inspection of concrete construction whenever inspection services are required by the local building codes or the Contract Documents. The services covered are directed to Special Inspections covered in the current edition of the IBC. This specification is also valid for other projects at the option of the user. It includes inspection requirements for preplacement (reinforcement, forms, and embedments), placement (conveyance and consolidation), and post-placement (finishing, curing, stressing, grouting, and formwork removal) of concrete construction, as well as requirements for personnel qualification and Inspection Agency accreditation. All inspection services required in this specification shall be performed by an Inspection Agency engaged by the Owner.

1.1.2 Independence—An Inspection Agency shall be objective, competent, and independent from the Contractor responsible for the Work being inspected. The Owner or the Owner's Representative shall not delegate these services to the Contractor, Construction Manager, Construction Manager at Risk, or any other party responsible for constructing the Work.

1.1.3 Coordination with Contractor—The Owner shall instruct the Contractor to coordinate with the Inspection Agency, provide access to the Work, and provide notification when the Work is ready for inspection.

1.2—Units of measurement

1.2.1 Units—Inch-pound is the standard units of measurement in this document.

1.3—Referenced standards

Standards cited in this specification are listed by name and designation, including year, unless otherwise specified.

1.3.1

ACI 301-16—Specifications for Structural Concrete

ACI 311.6-18—Specification for Ready Mixed Concrete Testing Services

ACI 318-14—Building Code Requirements for Structural Concrete

ACI 355.4-11—Qualification of Post-Installed Adhesive Anchors in Concrete and Commentary

1.3.2 ASTM International

ASTM A706/A706M-15—Standard Specification for Deformed and Plain Low-Alloy Steel Bars for Concrete Reinforcement

ASTM E329-14—Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection

1.3.3 American Welding Society

AWS D1.4/D1.4M: 2011—Structural Welding Code—Reinforcing Steel

1.3.4 International Code Council

IBC 2015

1.4—Definitions

The following definitions govern in this specification. For definitions of other terms, refer to “ACI Concrete Terminology (ACI CT).”

acceptance—acknowledgement by Architect/Engineer that submittal or completed work is acceptable.

accreditation—act or result of having an Inspection Agency's quality system, including equipment calibration and personnel certifications, reviewed and accredited by a qualified third-party Accreditation Agency.

Architect/Engineer—architect, engineer, architectural firm, or engineering firm developing Contract Documents or administering the Work under construction Contract Documents, or both.

certified—having met the requirements of a program that determines and attests that an individual is qualified to perform a specific service.

Contractor—person, firm, or entity under contract for the construction of the Work.

continuous inspection—full-time observation of Work requiring inspection by a Special Inspector who is present in the specific area where the Work is being performed.

discrepancy—an item that does not conform to the contract documents and is reported to the contractor for correction.

inspection agency—person, firm, or entity under contract for providing inspection services.

inspection services—periodic or continuous Special Inspections as required in this specification and Inspection Services Contract between the Owner and Inspection Agency.

Inspection Services Contract—set of documents supplied by the Owner to the Inspection Agency that serves as the basis for the agreement to perform inspection services; these documents can contain contract forms, contract conditions, Statement of Special Inspections, specifications, drawings, addenda, and contract changes. A different set of Contract Documents will exist between the Owner and construction Contractor.

Non-conformance—discrepancy that is not corrected prior to completion of that phase of the work and is reported to the building official and RDPRC so the RDPRC can determine the action to be taken.

Owner—corporation, association, partnership, individual, or public body or authority for whom the Work is constructed.

periodic inspection—part-time or intermittent observation of Work requiring Special Inspection by a Special Inspector who is present in areas where the work is being performed or at the completion of the Work. All items requiring inspection must be inspected before closing or before the Work is no longer accessible to inspect.

Project Specification—written document that details requirements for the Work in accordance with service parameters and other specific criteria.

Registered Design Professional in Responsible Charge—used in place of Architect/Engineer in the IBC.

Statement of Special Inspection—a list of the required special inspections and code references that is based on the project contract documents and prepared by the RDPRC.

Special Inspection—inspection of construction requiring the expertise of a Special Inspector to ensure compliance with the approved Contract Documents.

Special Inspector—employee of the Special Inspection Agency or an individual that is qualified and certified to perform the inspections described in this specification

testing agency—person, firm, or entity under contract for monitoring the quality of the materials as specified in the Contract Documents.

verify—act of reviewing, inspecting, testing, and reporting that Work inspected is or is not in conformance with Contract Documents

Work—entire construction or separately identifiable parts thereof required to be furnished under the Contract Documents.

1.5—Reporting requirements

1.5.1 Comply with special inspection requirements of the applicable Building Code and with the additional requirements of the Contract Documents.

1.5.2 Notify the Contractor of discrepancies from approved Contract Documents daily, usually as soon as detecting the discrepancy. If uncorrected prior to the completion or covering of the Work, before further Work occurs, notify the Contractor, Owner, Architect/Engineer, and building official with copies of the Non-Conformance Report (NCR).

1.5.3 Compile an NCR log (NCL) or similar document noting construction items not conforming to the Contract Documents. The NCL shall describe the nonconformance, date noticed, and individuals or organizations informed. Upon completion of the corrective action, update the NCR and NCL to indicate the action and date of acceptance by the Architect/Engineer, Building Official, or both, as appropriate.

1.5.4 Submit progress reports to the Owner, Architect/Engineer, and the Building Official, describing inspections and tests that were performed, nonconformances, and compliance of Work with the Contract Documents.

1.5.5 Submit a final summary report stating whether Work requiring inspection was, to the best of the inspector's knowledge, in conformance with the approved Contract Documents and applicable provisions of the building code.

1.6—Qualifications

1.6.1 *Inspection agency*—Unless otherwise specified, Inspection Agency shall submit to Owner its qualifications and the qualifications of its inspectors as required in 1.6.1.1 and 1.6.1.2.

1.6.1.1 *Special Inspectors*—Unless otherwise specified, Inspectors conducting Special Inspections of concrete preplacement (reinforcement, embedments, and forms), placement (conveyance and consolidation), and post-placement (finishing, curing, stressing, grouting, and form and shore removal) as defined in the local building code, shall be currently certified by ACI as a Concrete Construction Special Inspector (CCSI) or Concrete Transportation Construction Inspector (CTCI) or as acceptable to the local building official.

1.6.1.2 *Inspection Agency*—Unless otherwise specified, The Inspection Agency shall conform to the requirements of **ASTM E329** as it pertains to concrete inspection and local building code requirements. The Inspection Agency shall be accredited by The American Association for Laboratory Accreditation (A2LA) or The International Accreditation Service (IAS) or as acceptable to the local Building Official.

1.7—Items requiring inspection

1.7.1 Unless otherwise specified in the local building code or Contract Documents, items requiring verification and inspection shall be continuously or periodically inspected in accordance with Table 1.7.1. Any item not listed in Table 1.7.1 shall be deemed requiring periodic inspection.

SECTION 2—PREPLACEMENT INSPECTION

All work listed in Section 2 shall be verified in accordance with Table 1.7.1 and **ACI 301** Section 4, and as specified in the Contract Documents.

2.1—General requirements

2.1.1 Review the Contract Documents and required local Building Codes for inspection requirements.

2.1.2 Review structural and material submittals accepted by the Architect/Engineer or RDPRC.

2.1.3 Take part in preconstruction meetings to discuss project inspection requirements and proper sequencing of required inspection services, and the duties of all parties concerned.

2.1.4 Before concrete construction, verify that the substrate condition (soil subgrade, decking, or other materials beneath the concrete construction) has been inspected and approved for concrete placement by the Architect/Engineer or RDPRC.

2.1.5 Verify use of required design mixture per Table 1.7.1 Item 5.

2.2—Formwork, shoring, and embedments

All work listed in 2.2 shall be verified per Table 1.7.1 Item 12; **ACI 301**, 2.3; and as specified in the Contract Documents.

Table 1.7.1—Required Special Inspections of concrete construction

Item	Continuous special inspection	Periodic special inspection	Referenced standard and codes
1. Inspect reinforcement, including prestressing tendons, and verify placement.	—	X	ACI 318, Ch. 20, 25.2, 25.3, and 26.6.1 through 26.6.4
2. Reinforcing bar welding:			
a) Verify weldability of reinforcing bars other than ASTM A706	—	X	AWS D1.4;
b) Inspect single-pass fillet welds, maximum 5/16 in.	—	X	ACI 318, 26.6.4
c) All other welds	X	—	
3. Inspect anchors cast in concrete.	—	X	ACI 318, 17.8.2
4. Inspect anchors post-installed in hardened concrete members.			
a) Adhesive anchors installed in horizontally or upwardly inclined orientations to resist sustained tension loads.	X	—	ACI 318, 17.8.2.4; ACI 355.4
b) Mechanical anchors and adhesive anchors not defined in 4a.	—	X	ACI 318, 17.1.2 and 17.8.2
5. Verify use of required design mixture.	—	X	ACI 318, Ch. 19 and 26.4.4.1
6. During concrete and shotcrete placement, verify fabrication of strength test specimens and perform slump, air content tests, and determined temperature of the concrete.	X	—	ACI 311.6; ACI 318, 26.12
7. Inspect concrete placement for proper application techniques.	X	—	ACI 318, 26.5.2
8. Verify maintenance of specified curing temperature and techniques.	—	X	ACI 318, 26.5.3.2 through 26.5.5.2
9. Inspect prestressed concrete for:			
a) Application of prestressing forces	X	—	ACI 318, 26.10.1 and
b) Grouting of bonded prestressing tendons.	X	—	26.13.3.2
10. Inspect erection of precast concrete members.	—	X	ACI 318, 26.9
11. Verify in-place concrete strength prior to stressing of tendons in post-tensioned concrete and prior to removal of shores and forms from beams and structural slabs.	—	X	ACI 318, 26.11.2.1 and 26.13.3.3e
12. Inspect formwork for shape, location, and dimensions of the concrete member being formed.	—	X	ACI 318, 26.11.1.2(b)
13. Inspect shotcrete placement for proper application techniques, materials, and testing	X	—	IBC 1908

2.2.1 Verify the formwork complies with requirements for cleanliness and will provide concrete elements of the specified size and shape.

2.2.2 Verify that the location and preparation of joints comply.

2.2.3 Verify that the type, quantity, size, spacing, and location of embedments are as specified and are supported and secured with materials approved by the Architect/Engineer or RDPRC, to prevent displacement during concrete placement.

2.2.4 Verify that block-outs are correct for location, size, and condition.

2.3—Mild steel reinforcement

All work listed in 2.3 shall be verified per Table 1.7.1 Items 1 and 2; **ACI 301**, 3.3; and as specified in the Contract Documents and approved placing drawings.

2.3.1 Verify that reinforcing steel is of the type, grade, and size specified.

2.3.2 Verify that reinforcing steel is free of oil, dirt, and loose rust and that steel is acceptably coated, if specified.

2.3.3 Verify that reinforcing steel is located in accordance with the Contract Documents and is adequately supported and secured with materials approved by the Architect/Engineer or RDPRC, to prevent displacement during concrete placement.

2.3.4 Verify that minimum concrete cover will be provided based on formwork configuration.

2.3.5 Verify that placement of reinforcing steel complies with required spacing, profile, and quantity requirements, as indicated in the Contract Documents.

2.3.6 Verify that hooks, bends, ties, stirrups, and supplemental reinforcements are fabricated and properly placed.

2.3.7 Verify that required lap lengths, stagger, and offsets are provided.

2.3.8 Verify proper installation of approved mechanical connections in accordance with the manufacturer's instruction, evaluation reports, or both.

2.3.9 Verify that welds of reinforcing steel and other elements are as specified and have been inspected and approved by a certified welding inspector.

2.4—Bonded post-tensioned steel reinforcement

All work listed in 2.4, 2.5, and 2.6 shall be verified per Table 1.7.1 Items 7, 9, and 11; **ACI 301**, 13.2; and as specified in the Contract Documents.

2.4.1 Verify that the handling and storage of tendon strands on-site did not contaminate or damage strands.

2.4.2 Verify placement and condition of tendons and anchorages.

2.4.3 Verify that ducts, including inlets and outlets, are of the required size, mortar tight, and are in specified locations.

2.4.4 Verify that the pump and jack system to be used for tendon tensioning has been calibrated within the prior 6 months.

2.4.5 Verify that the prestressing force is applied within the allowable tolerance and is recorded accurately.

2.4.6 Verify that the stressing sequence is followed, if specified.

2.4.7 Verify that the elongations are within the required tolerance and are recorded accurately.

2.5—Unbonded post-tensioned steel reinforcement

2.5.1 Verify that the handling and storage of tendon strands on-site did not contaminate or damage strands.

2.5.2 Verify placement location of tendons and anchorages and that sheathing is intact.

2.5.3 Verify that procedure approved by the Architect/Engineer or RDPRC is followed for any repair of sheathing.

2.5.4 Verify location and profile of tendons and ducts.

2.5.5 Verify that tie bars, chairs, and bolsters are the specified type and are secured to the tendon to prevent displacement of tendon during concrete placement.

2.6—Stressing and grouting

2.6.1 Verify that the pump and jack system to be used for tendon tensioning have been calibrated within the prior 6 months.

2.6.2 Verify that the stressing sequence is followed, if specified.

2.6.3 Verify that the stressing forces and tendon elongations are within the required tolerances and are recorded accurately.

2.6.4 Verify that the in-place concrete strength is achieved before transferring stressing force or stressing.

2.6.5 Verify that the grout mixture for tendon grouting is approved by Architect/Engineer or RDPRC, the grout is applied continuously until the vents discharge a steady stream of grout, and the grouting pressure is attained after the vents are closed.

SECTION 3—PLACEMENT INSPECTION

All work listed in Section 3 shall be verified per Table 1.7.1 Items 6 and 7; **ACI 301**, 2.3 and 5.3; tested per **ACI 311.6**; and as specified in the Contract Documents.

3.1—Concrete quality

3.1.1 Verify that all individual batch tickets indicate deliveries of the mixture are as specified.

3.1.2 Verify time limits of mixing, total water added, drum revolution limits and specified consistency, and workability for the placements are as specified.

3.1.3 Verify that the required type, quantity, and frequency of tests to be performed on the fresh concrete are performed

by a certified ACI Concrete Field Testing Technician, as specified in ACI 311.6.

3.1.4 Observe sampling of concrete, field testing of fresh concrete, and making of test specimens.

3.1.5 Verify slump, temperature, and air content.

3.2—Conveyance and placement

3.2.1 Verify acceptable condition of the place of deposit before the concrete is placed. Place of deposit shall be free of snow, ice, mud, water, and loose material.

3.2.2 Verify that methods of conveying and depositing concrete avoid contamination and segregation of the mixture.

3.2.3 Verify use of drop chutes or funnel hoses to contain free fall and that free fall does not exceed requirements.

3.2.4 Verify that grout used to lubricate the pump hose is not incorporated into the placement.

3.2.5 Report any unexpected delays that may have caused a cold joint.

3.2.6 Verify that the depths of concrete layers are as specified.

3.3—Consolidation

3.3.1 Verify that the vibrator is inserted through the full depth of the layer and 6 in. into the previous layer and that the vibrator is not used to move concrete horizontally.

3.3.2 Verify that the spacing and duration of external vibration are consolidating the concrete without segregation.

3.3.3 Verify that revibration is performed if required.

3.3.4 Verify that formwork is tight and appears stable.

3.4—Finishing

3.4.1 Verify that screeding, floating, edging, troweling, and texturing are completed.

3.4.2 Verify that saw-cut joints are cut within the specified time and within tolerances for depth, width, location, and alignment.

SECTION 4—POST-PLACEMENT INSPECTION

All work listed in Section 4 shall be verified per Table 1.7.1 Item 8; **ACI 301**, 5.3.6; and as specified in the Contract Documents.

4.1—Protection and curing

4.1.1 Verify that concrete is protected from temperature extremes and from sunlight and wind that could detrimentally decrease the quality of the hardened concrete or produce plastic shrinkage cracking due to rapid drying, as in accordance with the Contract Documents.

4.1.2 Verify that the curing compound or method approved by the Architect/Engineer or RDPRC, if specified, is applied at the specified time and rate of application.

4.1.3 Verify that the specified curing is applied and for the specified duration, per the Contract Documents.

4.2—Forms, shore removal, and reshoring

4.2.1 Verify that form removal, shore removal, and reshoring are performed per the Contract Documents.

4.2.2 Verify that the concrete has attained the minimum required in-place strength before formwork removal by reviewing the strengths of field-cured specimens or the results of in-place test obtained by testing agency.

4.2.3 Observe and document the condition of the concrete surface after form removal and report to the Architect/Engineer or RDPRC on the possible need for repairs.

4.3—Strength test specimens

4.3.1 Verify that proper test specimen identification is used and that the correct numbers of specimens are cast.

4.3.2 Verify the initial site storage, curing, and protection of strength test specimens are per contract documents.

4.3.3 Verify that specimens are protected against damage while stored, and loaded into transportation vehicle to the Testing Laboratory.

4.3.4 Verify the storage conditions and locations of field-cured test specimens used to confirm in-place concrete strength before form and shoring removal and tendon stressing or transfer of tendon stress.

4.3.5 Verify that the strength of test specimens is achieved in accordance with the Contract Documents by review of the testing agency reports.

4.3.6 Verify the testing agency conforms to **ACI 311.6**, unless otherwise specified.

SECTION 5—CONCRETE ANCHORS

All work listed in Section 5 shall be verified per Table 1.7.1 Items 3 and 4; **ACI 355.4**; and as specified in the Contract Documents.

5.1—General

5.1.1 Verify concrete has reached an age of 21 days before installation of anchors.

5.1.2 Verify that the installer for adhesive anchors is a certified ACI Adhesive Anchor Installer, unless otherwise specified.

5.1.3 Verify installer has the proper manufacturer's product installation instructions (MPII).

5.1.4 Verify the installer has a copy of the design requirement for the field condition of the concrete, including minimum age of concrete; concrete temperature range; moisture condition of concrete at time of installation; type of lightweight concrete, if applicable; and requirements for hole drilling and preparation.

SECTION 6—ERECTION OF PRECAST CONCRETE

All work shall be verified per Table 1.7.1 Item 10; **ACI 301**, 13.3; and as specified in the Contract Documents.

SECTION 7—SHOTCRETE

All work shall be verified per Table 1.7.1 Item 13, and as specified in the Contract Documents.

(nonmandatory section follows)

NOTES TO SPECIFIERS

General notes

G1. ACI Specification 311.7-16 is to be used by reference or incorporation in its entirety in the Project Specification. Do not copy individual parts, sections, articles, or paragraphs into the project specification because taking them out of context may change their meaning.

G2. If sections or parts of ACI Specification 311.7-16 are copied into the Project Specification or any other document, do not refer to them as an ACI Specification, because the specification has been altered.

G3. A statement such as the following will serve to make ACI Specification 311.7-16 a part of the Project Specification:

“Work on (Project Title) shall conform to all requirements of ACI Specification 311.7-16, ‘Specification for Inspection of Concrete Construction,’ published by the American Concrete Institute, Farmington Hills, MI, except as modified by these contract documents.”

G4. Each technical Section of ACI Specification 311.7-16 is written in the three-part Section format of the Construction Specifications Institute, as adapted for ACI requirements. The language is imperative and terse.

G5. If ACI Specification 311.7-16 is used with another ACI specification that contains overlapping provisions, identify which requirements are in conflict and state in the contract documents which requirements control.

Foreword to checklists

F1. This foreword is included for explanatory purposes only; it is not a part of ACI Specification 311.7-16.

F2. ACI Specification 311.7-16 may be referenced by the specifier in the inspection services contract for any building project, together with supplementary requirements for the specific project. Responsibilities for project participants must be defined in the inspection services contract. ACI Specification 311.7-16 cannot and does not address responsibilities for any project participant other than inspection agency.

F3. Checklists do not form a part of ACI Specification 311.7-16. Checklists assist the specifier in selecting and specifying project requirements in the inspection services contract.

F4. The Optional Requirements Checklist identifies specifier choices and alternatives. The checklist identifies the sections, parts, and articles of ACI Specification 311.7-16 and the action required or available to the specifier. The specifier should review each of the items in the checklist and make adjustments to the needs of a particular project by including those selected alternatives as mandatory requirements in the inspection services contract.

OPTIONAL REQUIREMENTS CHECKLIST

Section/Part/Article	Notes to Specifiers
1.3	References other than the latest edition of the included documents may be specified.
1.6.1 and 1.6.1.1	Other special inspectors' certifications equivalent to ACI CCSI or CTCI may be specified if allowed by the building code official. Equivalent certification should include written and performance examinations to relevant codes and specifications as accepted by the Accreditation Agency.
1.6.1.2	Inspection agency accreditations equivalents may be specified.
1.7.1	Inspection may be changed from continuously or periodically, by the local building codes.
4.2.6	Contractors' testing agency accreditations' equivalents may be specified.
5.1.2	Anchor installer certification equivalents may be specified.



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As ACI begins its second century of advancing concrete knowledge, its original chartered purpose remains “to provide a comradeship in finding the best ways to do concrete work of all kinds and in spreading knowledge.” In keeping with this purpose, ACI supports the following activities:

- Technical committees that produce consensus reports, guides, specifications, and codes.
- Spring and fall conventions to facilitate the work of its committees.
- Educational seminars that disseminate reliable information on concrete.
- Certification programs for personnel employed within the concrete industry.
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Individuals interested in the activities of ACI are encouraged to explore the ACI website for membership opportunities, committee activities, and a wide variety of concrete resources. As a volunteer member-driven organization, ACI invites partnerships and welcomes all concrete professionals who wish to be part of a respected, connected, social group that provides an opportunity for professional growth, networking and enjoyment.

